

Vehicle Type-Based Alert System for Hazardous Pathways

A Machine Learning-Based Real-Time Accident Prediction System for Hill Country Roads

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This document summarises the submitted research proposal and the implementation progress completed so far: data collection, preprocessing, feature engineering, model training, explainability, and prototype system development.

Source basis: submitted proposal PDF, project README, processed datasets, preprocessing report, model comparison report, RF tuning report, SMOTE outputs, SHAP report, and generated visualisations.

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1. Executive Summary

The project aims to develop a vehicle-specific real-time accident risk prediction and alert system for mountainous roads, with Kadugannawa selected as the primary study area. The core idea is that cars, buses, lorries, motorcycles, vans, three-wheelers, and pick-ups do not experience the same risk under the same road and weather conditions. A steep sharp bend during rain is more hazardous for a loaded lorry or motorcycle than for a small car, so the alert system must consider vehicle type together with weather, road geometry, time, driver, and telemetry conditions.

The proposal originally positioned Random Forest as the primary machine learning model because it can model non-linear patterns, handle mixed feature types, and provide interpretable feature importance. The implementation has now expanded the modelling stage to include Random Forest tuning, SMOTE-based imbalance handling, XGBoost comparison, and SHAP explainability. This strengthens the academic contribution because it allows the final study to report both the proposed model and a stronger comparative baseline.

Current status: A complete research prototype pipeline exists. It includes mock police accident records, real weather data, OpenMeteo API integration, OpenStreetMap road geometry extraction, preprocessing, feature engineering, Random Forest and XGBoost training, model evaluation, SHAP explanation, backend API structure, frontend dashboard pages, and an edge-device telemetry simulator.

2. Alignment with Submitted Proposal

Proposal Component	Proposal Requirement	Implementation Completed So Far
Research topic	Vehicle type-based alert system for hazardous pathways using ML for hill country roads.	Project structure and implementation are built around Kadugannawa mountain-road accident risk prediction.
Historical data	Police or insurance accident records containing date/time, vehicle type, location, weather, and accident cause.	Created 500-row mock police accident dataset and included <code>Police.records.data.Kadugannawa.csv</code> with the same structure for later real-data replacement.
Real-time weather	Weather API for current environmental conditions.	Implemented OpenMeteo collection scripts and generated historical and forecast CSV files for Kadugannawa.
Topographical data	Road gradient, elevation, and curve sharpness.	Implemented OpenStreetMap road geometry extraction and produced Kadugannawa road segment features.
Machine learning	Random Forest classifier with train-test split, cross-validation, and tuning.	Implemented RF baseline, RF + SMOTE, tuned RF, XGBoost comparison, cross-validation, and threshold analysis.
Evaluation	Accuracy, precision, recall, F1-score, confusion matrix, and feature importance.	Generated comparison reports, ROC curves, confusion-matrix outputs, feature importance plots, and SHAP explanations.
System development	Backend prediction engine and frontend/mobile dashboard for real-time alerts.	Backend FastAPI-style modules, frontend dashboard pages, weather API endpoints, prediction endpoint, and edge telemetry simulator have been scaffolded.

3. Research Problem and Research Questions

Problem Statement

Mountainous roads such as Kadugannawa contain steep gradients, sharp curves, sudden weather changes, and visibility limitations. Existing accident prediction systems are mainly designed for urban or highway environments, often treat all vehicles equally, and rarely combine historical accident patterns with real-time weather and terrain conditions. Therefore, there is a need for a vehicle-specific, terrain-aware machine learning model that can provide early accident risk alerts.

Research Questions

1. What machine learning algorithms have been used for early road accident prediction?
2. What road-safety factors in mountainous areas and vehicle-specific factors influence accident risk?
3. How can an accurate machine learning algorithm be developed to identify accident risk in advance within mountainous environments?
4. How can the algorithm be evaluated to ensure it supports safer driving through early warnings?

4. Research Goal and Objectives

The main objective is to develop a vehicle-specific real-time accident risk prediction and alert system for mountainous roads so that drivers can receive early warnings and make safer decisions.

Objective	Description	Current Progress
RO1	Review ML accident prediction algorithms and identify mountain-road and vehicle-specific risk factors.	Proposal literature review completed; feature set now reflects weather, road geometry, vehicle, time, driver, and telemetry risk factors.
RO2	Develop an accurate ML algorithm using historical data to identify accident risk in advance.	Random Forest, tuned Random Forest, SMOTE variants, and XGBoost comparison implemented.
RO3	Evaluate the accuracy and performance of the developed algorithm using real-time data.	Hold-out evaluation, cross-validation, ROC-AUC, F1-macro, class-wise F1, SHAP, and feature importance outputs generated. Live API integration is partially implemented and needs real deployment validation.

5. Methodology

The methodology follows the proposal workflow but has been converted into an executable Python project. The current implementation uses a modular pipeline so each stage can be replaced or improved independently when real police records become available.

1. Data Collection Accident, weather, road, and telemetry data.	2. Preprocessing Cleaning, merging, missing-value handling, and leakage prevention.	3. Feature Engineering Vehicle, weather, road, time, speed, and interaction features.	4. Modelling RF, RF + SMOTE, tuned RF, and XGBoost.
5. Evaluation F1-macro, class-wise F1, recall, precision, kappa, AUC-ROC.	6. Explainability Feature importance and SHAP interpretation.	7. System Prototype Backend API, dashboard, and device telemetry simulator.	8. Validation Real police data replacement and field-style testing next.

6. Data Sources and Dataset Status

Dataset	File / Source	Status	Purpose
Mock accident records	data/raw/kadugannawa_accidents_mock.csv	500 records generated	Prototype training while waiting for official police data.
Police data placeholder	data/raw/Police.records.data.Kadugannawa.csv	Available in project	Structured so future real police records can replace mock data with minimal pipeline change.
Weather export	data/raw/export.xlsx	Available	Historical Kadugannawa weather attributes such as temperature, precipitation, wind, and pressure.
OpenMeteo weather	data/external/kadugannawa_historical_daily.csv, kadugannawa_forecast_7d.csv	Generated	Historical and forecast weather support for training and real-time alerts.
OpenStreetMap geometry	data/raw/kadugannawa_road_segments.csv	Generated	Road curvature, speed limit, slope, and location-based road segment features.
Telemetry sample	data/raw/telemetry/telemetry_raw.csv	Generated	Prototype speed, heading, acceleration, RPM, and GPS input for real-time risk scoring.

7. Data Preprocessing and Integration

The preprocessing pipeline merges accident, weather, road geometry, and telemetry-related features into a model-ready dataset. A key issue handled in the implementation is date alignment: accident dates and weather dates must be normalised before merging to prevent silent join failures. The pipeline also avoids leakage by removing outcome-derived columns such as injuries, fatalities, and manually generated risk scores before modelling.

Preprocessing Area	Implementation
Train-test split	Held-out test set retained before oversampling to avoid synthetic sample leakage.
Class imbalance	SMOTE applied to training data only. Final post-SMOTE training set in the preprocessing report: 510 rows, balanced as 170 Low, 170 Medium, 170 High.
Missing values	KNN and median imputers created for model features and stored for inference reuse.
Scaling	Scaler fitted only on training data; test set transformed using the fitted training scaler.
Output artifacts	<code>X_train.csv</code> , <code>X_test.csv</code> , <code>y_train.csv</code> , <code>y_test.csv</code> , <code>feature_names.csv</code> , and preprocessing report.

Important academic point: The primary metric should be F1-macro rather than accuracy because high-severity accidents are the minority class. A high accuracy score alone could hide poor High-risk detection.

8. Feature Engineering

Feature engineering expanded the dataset from raw observations into risk factors that better represent mountainous-road hazards. The feature registry ranks candidate features using mutual information and marks which features are used in the model.

Feature Group	Example Features	Reason for Inclusion
Weather	weather condition, visibility class, low visibility, fog risk, rainfall intensity	Fog, rain, low visibility, and pressure changes are critical in hill-country accident risk.
Road geometry	slope category, road curvature, sharp curve flag, steep road flag	Mountain roads introduce braking, turning, and stability risks not present on flat roads.
Vehicle type	vehicle risk weight, heavy vehicle flag, one-hot vehicle type features	Different vehicles respond differently to gradients, curves, rain, and visibility.
Speed and telemetry	speed ratio, speed excess, telemetry speed ratio, brake index	Real-time unsafe driving behaviour can raise alert level even in moderate road conditions.
Time	night, weekend, monsoon season, morning/evening peak	Accident patterns vary by visibility, traffic level, and seasonal rain exposure.
Interactions	rain x curve, visibility x night, slope x curve, lorry x steep road	Compound risk is non-linear: multiple moderate hazards together can create High risk.

Top Feature	Group	Mutual Information Score
weather_enc	Weather	0.197
location_risk	Location	0.195
slope_cat	Road	0.177
road_curv_enc	Road	0.171
is_sharp	Road	0.157
is_steep	Road	0.156
vis_class	Weather	0.147
slope_x_curve	Road interaction	0.140
high_risk_combo	Interaction	0.131
rain_x_curve	Interaction	0.119

9. Model Development and Evaluation

The proposal selected Random Forest as the main model. The implementation follows this and also compares XGBoost to strengthen the research evaluation. The main evaluation metric is F1-macro because it treats Low, Medium, and High severity classes equally and is more suitable for imbalanced safety data.

Model Comparison Results

Model	F1-Macro	Kappa	AUC-ROC	Accuracy
Random Forest default	0.8791	0.7817	0.9742	0.8700
Random Forest tuned	0.8791	0.7817	0.9736	0.8700
XGBoost default	0.8979	0.8298	0.9793	0.9000
XGBoost tuned	0.9120	0.8473	0.9797	0.9100

Class-Level Results for Tuned Models

Model	Low F1	Medium F1	High F1	High Precision	High Recall
Random Forest tuned	0.8254	0.8785	0.9333	1.0000	0.8750
XGBoost tuned	0.8852	0.9174	0.9333	1.0000	0.8750

The tuned XGBoost model currently performs best overall with F1-macro 0.9120. However, Random Forest remains valuable because it aligns with the proposal, is easier to explain to non-technical stakeholders, and can be used as the main interpretable baseline.

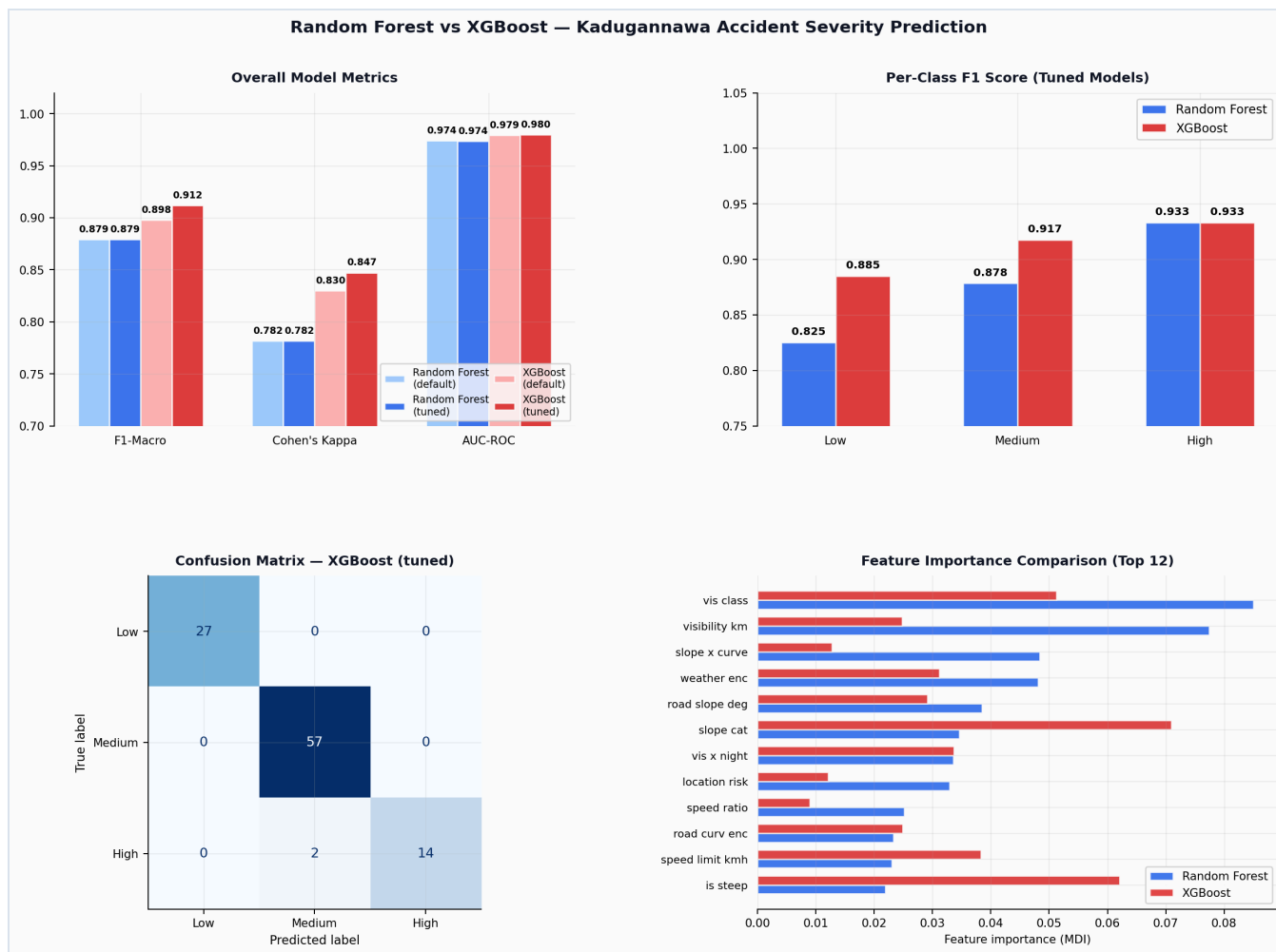


Figure 1. Model comparison visualisation generated from the experiment outputs.

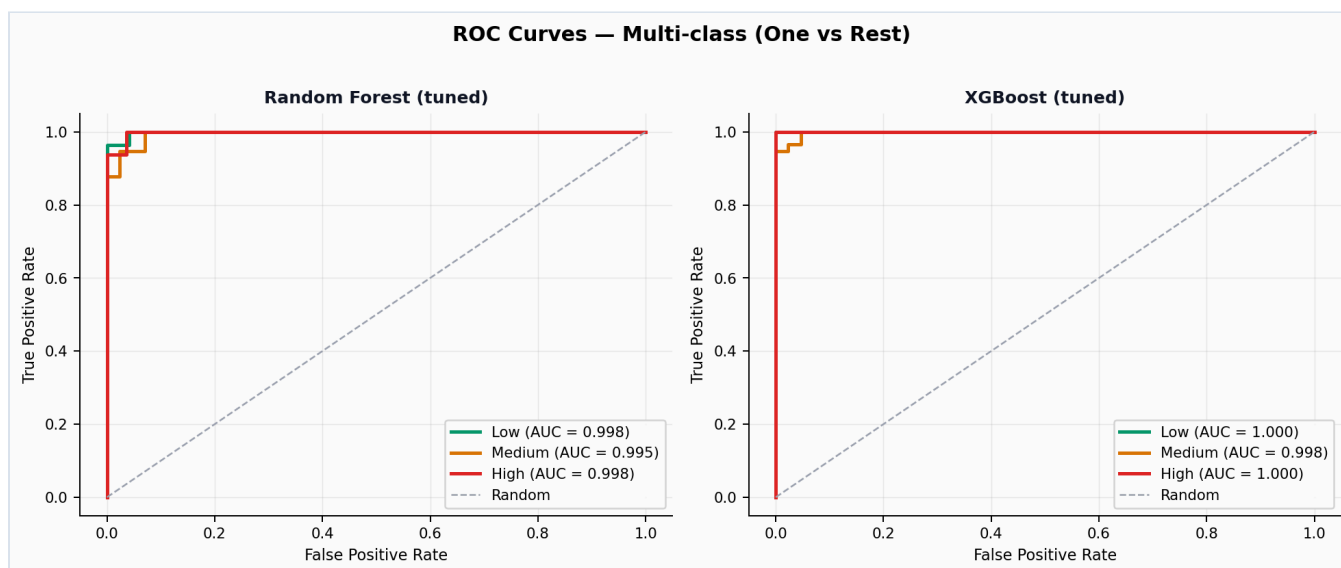


Figure 2. ROC curves for model evaluation.

10. Imbalanced Data Handling

Accident severity data is naturally imbalanced: severe accidents are less frequent but more important for safety. The project therefore tests imbalance-handling strategies such as class weighting, SMOTE,

and combined approaches. The RF tuning report shows that SMOTE improves the baseline Random Forest from F1-macro 0.8474 to 0.8815 after tuning.

Strategy	F1-Macro	High F1	High Precision	High Recall
Baseline, no handling	0.8472	0.8571	1.0000	0.7500
class_weight='balanced'	0.8474	0.8148	1.0000	0.6875
SMOTE only	0.8791	0.9333	1.0000	0.8750
SMOTE + balanced	0.8791	0.9333	1.0000	0.8750
SMOTE + balanced_subsample	0.8735	0.9032	0.9333	0.8750

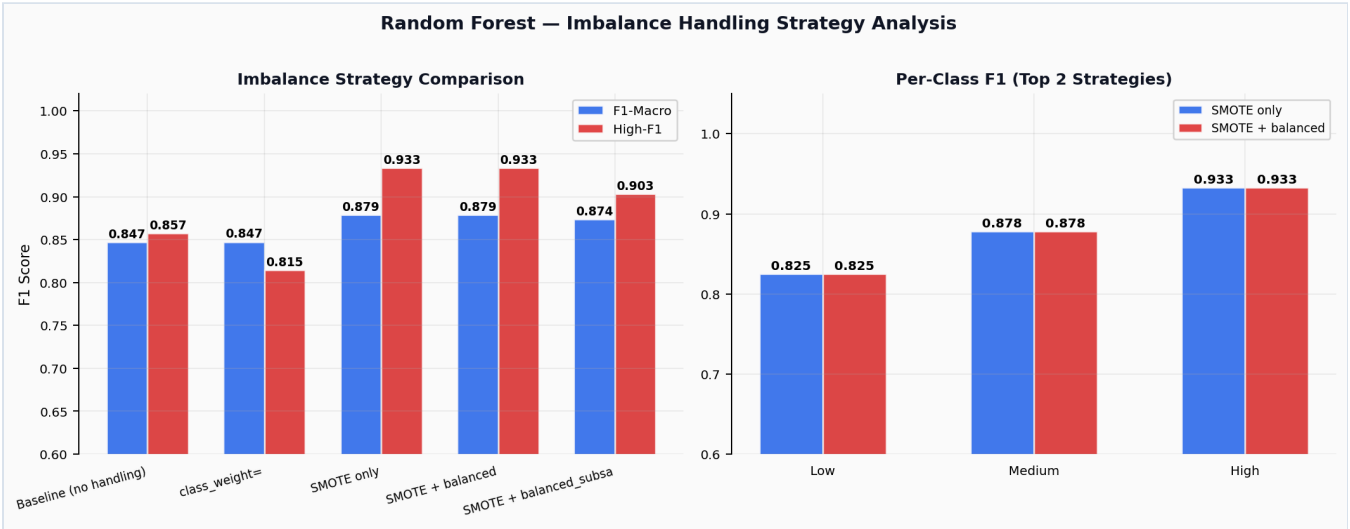


Figure 3. Random Forest imbalance-handling strategy comparison.

11. Explainability and Research Findings

SHAP analysis was applied to the tuned Random Forest model on the held-out test set. This is useful academically because it explains why the model predicts Low, Medium, or High risk. It also gives evidence for the research claim that weather, visibility, road geometry, and vehicle type interact in mountain-road accidents.

Finding 1: Weather and visibility dominate High severity. Visibility, visibility class, and weather condition are the strongest High-risk predictors in the SHAP report.

Finding 2: Road geometry compounds risk. Slope, curvature, and the slope x curvature interaction show that steep sharp bends create non-linear danger.

Finding 3: Vehicle type supports the proposal. Vehicle risk weight, lorry-steep interaction, and motorcycle-rain interaction support vehicle-specific risk modelling.

Finding 4: Time matters. Night-time and monsoon-season flags increase risk and support real-time contextual alerting.

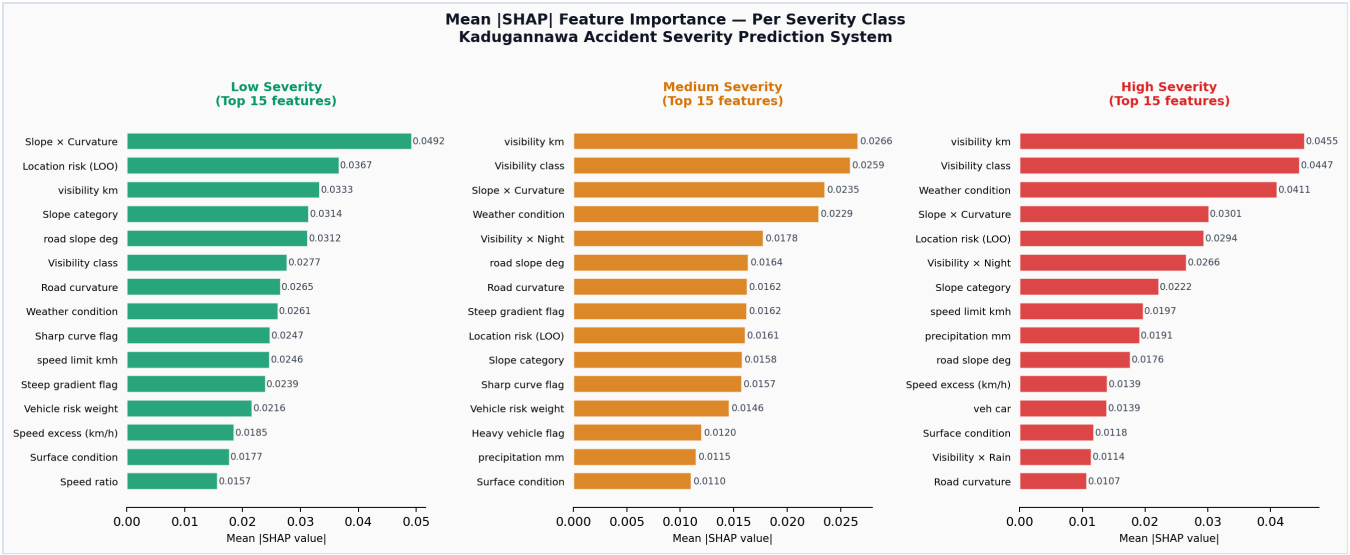


Figure 4. SHAP global feature importance.

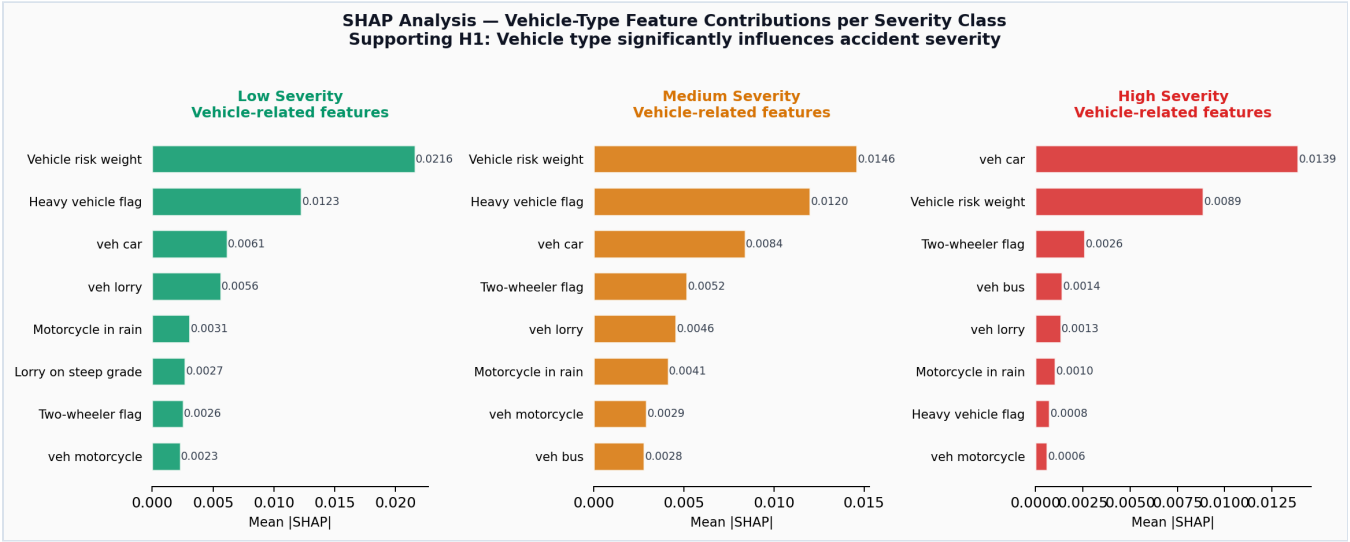


Figure 5. Vehicle-specific SHAP comparison supporting the vehicle-type research focus.

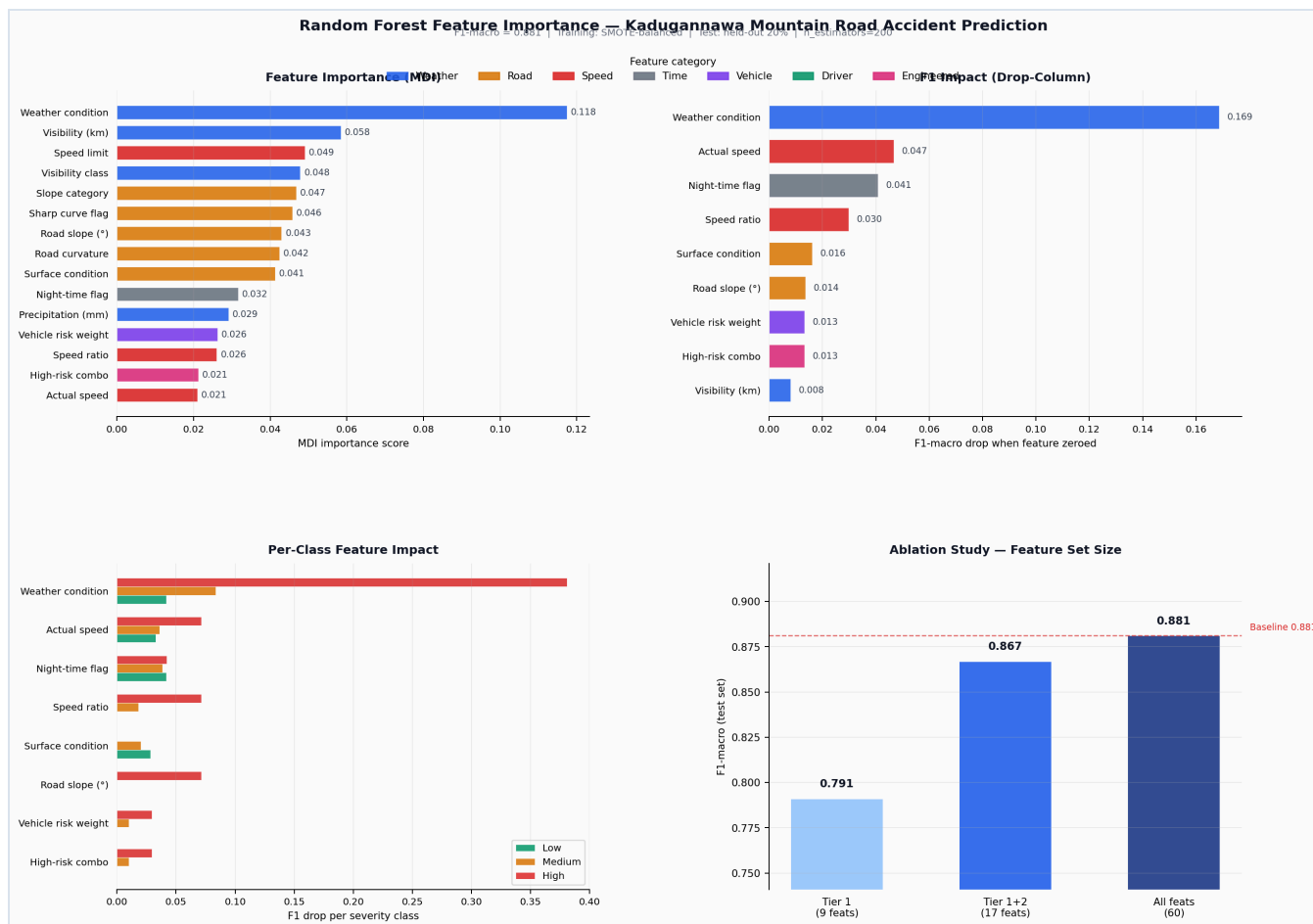


Figure 6. Feature importance visualisation for dissertation and supervisor discussion.

12. System Prototype

Beyond model training, the project has been organised as a deployable software system. The current architecture contains a backend, frontend, data processing pipeline, trained models, and an edge-device simulation layer.

Layer	Implemented Components	Purpose
Data pipeline	src/data_collection, src/preprocessing, src/feature_engineering	Collect weather/road data, merge records, and generate model-ready features.
Modelling	src/modeling, models/saved_models	Train, tune, compare, and save Random Forest and XGBoost models.
Evaluation	src/evaluation, outputs/shap, outputs/reports	Produce model comparison, feature importance, ROC, SHAP, and report files.
Backend	backend/api/predict.py, weather.py, segments.py, telemetry.py	Expose prediction, weather, road segment, and telemetry endpoints.
Frontend	frontend/public/dashboard.html, predict.html, JS components	Driver-facing dashboard and prediction UI.
Device layer	device/edge_agent.py, obd_adapter.py, telemetry queue	Simulate edge telemetry input such as speed, acceleration, and GPS.

The intended runtime flow is: driver selects or sends vehicle type, system fetches current weather, nearest road segment features and telemetry are added, the model predicts risk level, and the interface displays a Low, Medium, or High warning with vehicle-specific alert messaging.

13. Work Completed So Far

1. Extracted and reviewed the submitted proposal scope, problem statement, research questions, objectives, methodology, and expected system architecture.
2. Created a realistic 500-row mock Kadugannawa accident dataset to support development before official police data is available.
3. Integrated weather data using existing Excel weather records and OpenMeteo historical/forecast data.
4. Built OpenStreetMap road geometry extraction for slope, curvature, and road segment features.
5. Created preprocessing pipeline with date normalisation, left-join merging, imputation, train-test split, scaling, and leakage prevention.
6. Engineered 70+ candidate features covering weather, road, vehicle, speed, time, location, and interaction risks.
7. Applied SMOTE correctly to the training set only and compared imbalance-handling strategies.
8. Trained Random Forest baseline, tuned Random Forest, default XGBoost, and tuned XGBoost models.
9. Generated model comparison reports, ROC plots, feature importance plots, RF tuning plots, and SMOTE visualisations.
10. Implemented SHAP explainability and generated global, class-level, vehicle-level, and individual prediction explanations.

11. Scaffolded backend API, frontend dashboard, Docker configuration, and edge-device telemetry simulation.

14. Limitations and Research Risks

Risk	Impact	Mitigation
Mock accident data	Model scores may be optimistic because the current accident records are simulated.	Clearly state that current results are prototype results. Replace with official police data when available and rerun all experiments.
Class imbalance	High-severity accidents are rare, so accuracy alone can be misleading.	Use F1-macro, High-class recall, High-class precision, and SMOTE inside cross-validation.
Data leakage	Outcome variables such as injuries or fatalities could inflate performance.	Drop leakage columns before modelling and document this in the methodology.
Weather granularity	Daily weather may not represent accident-time conditions accurately.	Use OpenMeteo hourly or current API data for deployment and, if possible, accident-time matching.
Road geometry accuracy	OSM data may not perfectly represent local curve sharpness or surface condition.	Validate high-risk locations manually and compare with police/local authority knowledge.
Deployment validation	Prototype may behave differently with delayed APIs or weak network conditions.	Add caching, fallback weather values, offline road features, and input validation tests.

15. Next Steps Before Final Dissertation

1. Obtain and clean real police accident records for Kadugannawa or nearby mountain-road segments.
2. Map real accident locations to OSM road segments and validate GPS accuracy.
3. Replace mock dataset with real records and rerun preprocessing, feature engineering, model training, and SHAP analysis.
4. Use a dedicated validation set for threshold optimisation instead of choosing thresholds on the test set.
5. Finalise whether the dissertation will present Random Forest as the main model with XGBoost as comparison, or deploy XGBoost while retaining RF for interpretability.
6. Run backend API tests and demonstrate a real-time prediction request using current weather and selected vehicle type.
7. Prepare Chapter 4 methodology and Chapter 5 results using the generated reports and figures.
8. Add an ethics/data privacy section covering police data anonymisation, GPS privacy, and responsible alert design.

16. Suggested Supervisor Meeting Explanation

A concise way to explain the current progress:

I have implemented the full prototype pipeline for my proposed vehicle-type-based accident alert system. Since official police data is not yet finalised, I used a realistic Kadugannawa mock accident dataset with the same schema planned for replacement. I merged accident data with weather data, OpenMeteo API data, and OpenStreetMap road geometry, then performed feature engineering for vehicle, road, weather, time, speed, and interaction risks. I trained Random Forest as proposed and also compared it with XGBoost. The tuned XGBoost currently gives the best F1-macro score, while Random Forest remains the interpretable baseline. I also used SMOTE for imbalance handling and SHAP for explainability. The software prototype includes backend API modules, frontend dashboard pages, and telemetry simulation. My next key step is to replace mock accident records with real police data and rerun the full evaluation.

17. Questions to Ask the Supervisor

1. Should the final dissertation keep Random Forest as the primary model because it was proposed originally, or present XGBoost as the final deployed model with Random Forest as the baseline?
2. Is the current use of mock police data acceptable for prototype development until official records are received?
3. What exact validation evidence is expected for the final submission: model metrics only, prototype demo, user testing, or all three?
4. Should the scope include all vehicle types available in the dataset, or focus mainly on cars, motorcycles, buses, and lorries?
5. Is Kadugannawa alone sufficient as the case study, or should another mountain-road segment be added for comparison?

18. Appendix: Key Files Produced

Category	Files
Processed data	data/processed/merged_accidents_weather.csv, engineered_features.csv, X_train.csv, X_test.csv, feature_registry.csv
Models	models/saved_models/rf_model.pkl, rf_tuned_final.pkl, rf_smote_model.pkl, xgb_model.pkl
Reports	outputs/reports/comparison_report.txt, outputs/rf_tuning/tuning_report.txt, outputs/shap/shap_report.txt, data/processed/pipeline_report.txt
Figures	comparison_figure.png, roc_curves.png, feature_importance.png, shap_bar_global.png, shap_vehicle_comparison.png
System	backend/, frontend/, device/, docker-compose.yml